

ROLLA, MO, USA

WMO: 724420

Lat: 38.132N

Lon: 91.765W

Elev: 338

StdP: 97.33

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 13997

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.9	-11.8	-19.7	0.7	-13.5	-17.2	0.9	-10.5	11.6	5.8	10.7	4.5	3.7	310	0.515

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	34.8	24.0	33.0	23.9	31.6	23.6	25.8	31.6	25.1	30.9	24.4	29.8	3.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	19.8	29.5	23.4	19.0	28.5	22.7	18.2	27.6	81.7	31.6	78.5	30.8	75.4	29.9	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	8.9	8.0	-18.8	37.1	2.8	2.1	-20.8	38.6	-22.4	39.8	-23.9	41.0	-25.9	42.5		
(5)			-19.2	26.9	2.7	0.8	-21.1	27.5	-22.7	28.0	-24.2	28.4	-26.1	29.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.2	0.0	2.2	7.5	13.4	18.2	22.8	25.2	24.7	20.5	13.9	7.6	1.8		
	DBStd	10.20	6.74	6.36	6.16	5.24	4.45	3.36	2.99	3.14	4.13	5.29	5.71	5.96		
	HDD10.0	1084	314	229	121	25	2	0	0	0	0	22	110	261		
	HDD18.3	2600	567	453	337	162	59	6	1	1	25	156	322	512		
	CDD10.0	2261	6	11	44	127	257	383	472	456	314	144	39	7		
	CDD18.3	733	0	0	2	14	56	139	214	198	89	19	1	0		
	CDH23.3	6553	0	1	17	114	411	1209	2119	1812	737	126	7	0		
	CDH26.7	2547	0	0	1	18	93	426	943	792	253	21	0	0		
(14) Wind	WSAvg	4.0	4.6	4.6	4.7	4.8	3.9	3.5	3.1	3.1	3.3	4.0	4.4	4.5		
(15) Precipitation	PrecAvg	958	43	40	86	95	117	105	88	89	83	82	81	62		
	PrecMax	1332	131	102	181	201	256	316	264	270	183	156	230	153		
	PrecMin	725	6	7	18	28	43	2	23	19	12	5	17	11		
	PrecStd	188	35	27	41	43	53	80	60	60	45	44	58	39		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.1	21.7	25.5	28.9	31.2	34.4	37.4	37.3	33.7	29.1	24.3	19.4		
		MCWB	13.3	14.6	16.7	19.1	21.6	23.8	23.8	23.4	22.5	20.9	17.2	15.6		
	2%	DB	16.1	17.9	22.7	26.4	29.0	32.2	34.8	35.0	31.7	26.6	21.1	16.3		
		MCWB	11.8	12.8	15.3	17.5	21.7	24.2	24.3	24.0	22.7	19.5	14.6	12.9		
	5%	DB	12.8	15.3	20.0	24.2	27.5	30.8	33.0	33.0	29.7	24.4	18.6	13.5		
		MCWB	9.5	11.1	13.8	16.6	20.9	23.6	24.4	23.4	22.0	17.9	13.9	10.0		
	10%	DB	9.9	12.4	17.4	22.2	25.9	29.2	31.7	31.1	27.6	22.2	16.4	10.9		
		MCWB	6.5	8.5	12.4	15.7	19.8	22.9	24.1	23.1	20.9	17.1	12.2	7.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.5	16.2	18.2	20.5	24.0	26.2	26.8	26.7	24.7	22.5	18.4	16.6		
		MCDB	18.4	20.1	23.7	26.0	28.6	31.3	32.3	32.7	30.9	27.4	22.2	19.2		
	2%	WB	13.0	14.2	16.5	19.1	22.8	25.2	26.0	25.6	23.8	20.9	16.3	13.7		
		MCDB	15.0	17.3	20.9	24.0	27.5	30.5	31.8	31.5	29.5	24.4	19.5	15.8		
	5%	WB	9.9	11.6	15.0	17.9	21.7	24.2	25.3	24.7	22.9	19.6	14.8	10.8		
		MCDB	12.7	14.4	19.2	22.6	26.1	29.1	31.2	30.4	28.0	22.6	17.9	12.8		
	10%	WB	6.6	8.4	13.1	16.6	20.5	23.5	24.7	23.9	21.9	18.0	12.8	7.9		
		MCDB	9.7	12.0	16.8	21.3	24.6	28.2	30.1	29.2	26.3	21.4	16.0	10.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.9	10.2	11.2	11.6	10.8	10.4	11.0	11.2	11.4	11.0	10.5	9.3		
		MCDBR	13.5	14.5	13.9	13.6	12.0	11.5	12.6	13.3	12.5	12.3	12.7	12.7		
	5% WB	MCWBR	9.7	10.1	8.1	7.4	6.4	5.6	4.7	4.7	5.4	6.6	8.5	9.3		
		MCWBR	11.9	13.2	11.9	11.6	10.7	10.4	11.0	11.1	10.6	10.0	11.4	11.8		
(40) Clear-Sky Solar Irradiance	taub		0.297	0.309	0.335	0.381	0.417	0.424	0.449	0.448	0.385	0.342	0.307	0.294		
	taud		2.488	2.449	2.430	2.306	2.261	2.272	2.218	2.227	2.378	2.490	2.536	2.520		
	Ebn at Noon		890	923	928	898	866	857	832	824	862	873	872	870		
	Edh at Noon		79	94	106	127	136	134	141	136	110	89	75	72		
(44) All-Sky Solar Radiation	RadAvg		2.20	2.85	3.83	5.01	5.65	6.45	6.29	5.60	4.82	3.48	2.40	1.86		
	RadStd		0.24	0.31	0.36	0.31	0.50	0.50	0.42	0.30	0.32	0.38	0.25	0.20		

Historical Trends

		Heating		Cooling		Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (20 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
										+76

Nomenclature: See separate page